

ENERGY TRANSITION

Global Renewable Energy in 2025

How Far Have We Come?

A data-driven overview of renewables' growing share in global energy and electricity — hydropower, wind, and solar.

Data: Ember (2026) | Energy Institute Statistical Review of World Energy (2025)

Via: Our World in Data – ourworldindata.org/renewable-energy

Authors: Hannah Ritchie, Max Roser, Pablo Rosado

75% of Global GHG Emissions Still Come from Fossil Fuels

Setting the context: why the energy transition is urgent

75%

of GHG emissions
from fossil fuels

5M+

premature deaths/yr
from air pollution

~100%

fossil fuel dominance
since Industrial Revolution

Since the Industrial Revolution, fossil fuels have come to dominate most national energy systems. Burning coal, oil, and gas for energy is the **single largest driver of climate change** and generates severe local air pollution. Decarbonizing the energy system — through nuclear and renewable technologies — is essential to limiting warming and saving lives.

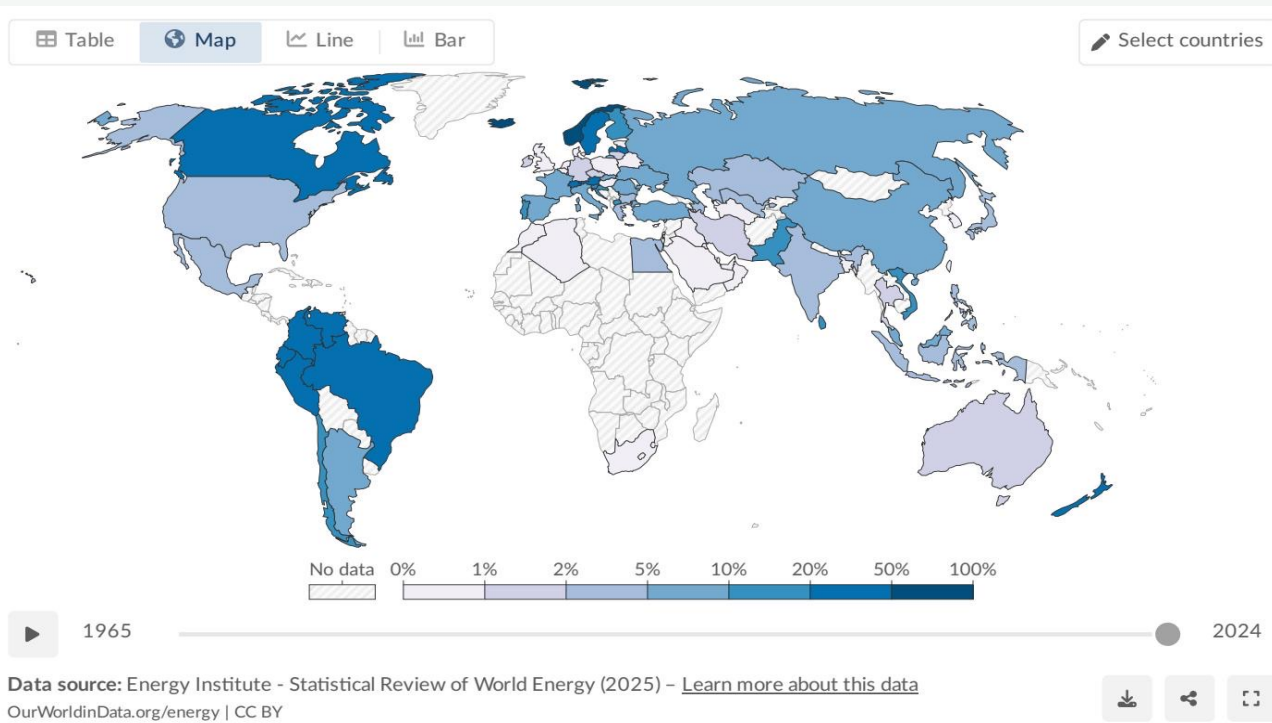
Renewables Now Supply ~14% of Global Primary Energy

2024 data (substitution method) — primary energy = electricity + transport + heating

~14%

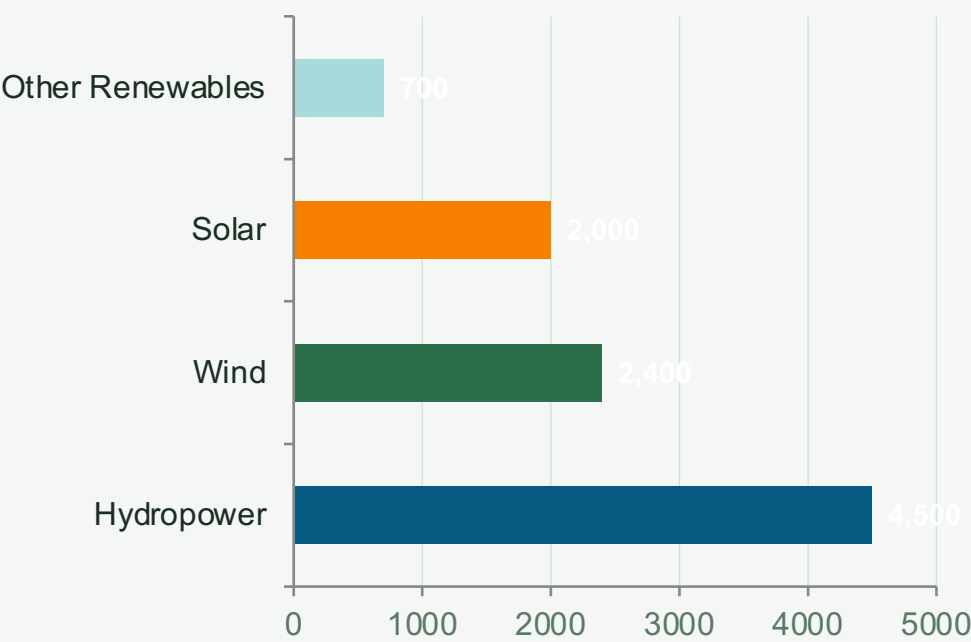
of global
primary energy
from renewables

One-seventh
of all energy
use is now
renewable



Renewable Electricity Generation: ~9,600 TWh Total in 2024

Hydropower leads — but wind and solar are growing fastest



Technology	TWh (2024)	% of Renewables
Hydropower	~4,500	~47%
Wind	~2,400	~25%
Solar	~2,000	~21%
Other (geo, biomass, etc.)	~700	~7%

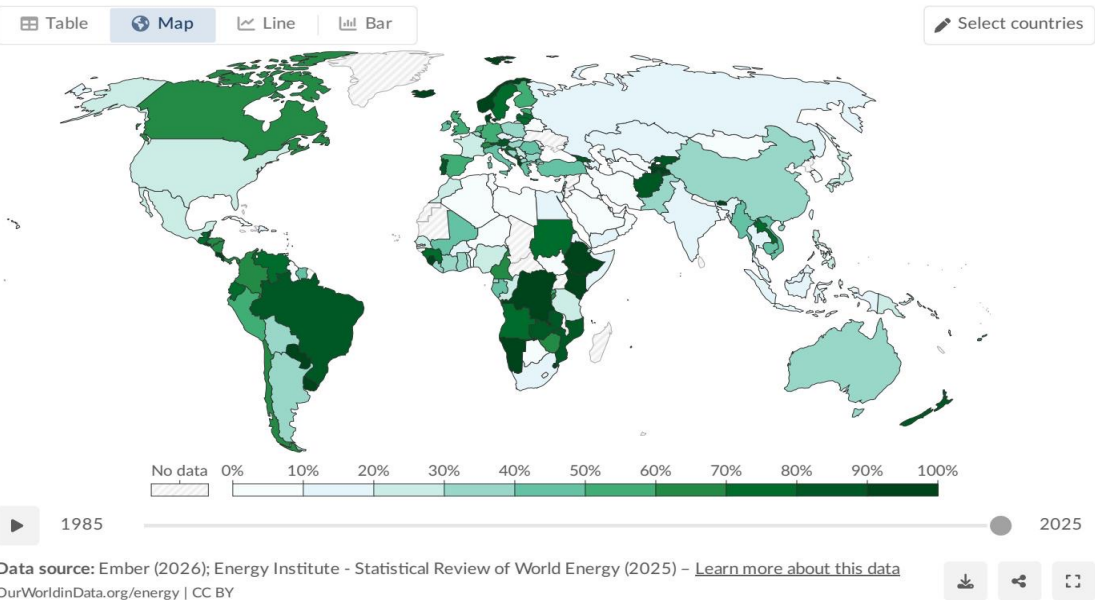
Wind + Solar combined (~4,400 TWh) now nearly match Hydropower's output and are growing far faster.

Almost One-Third of Global Electricity Now Comes from Renewables

2025 data — electricity sector leads all energy sectors in renewable deployment

Share of electricity production from renewables, 2025

Renewables include solar, wind, hydropower, bioenergy, geothermal, wave, and tidal sources.

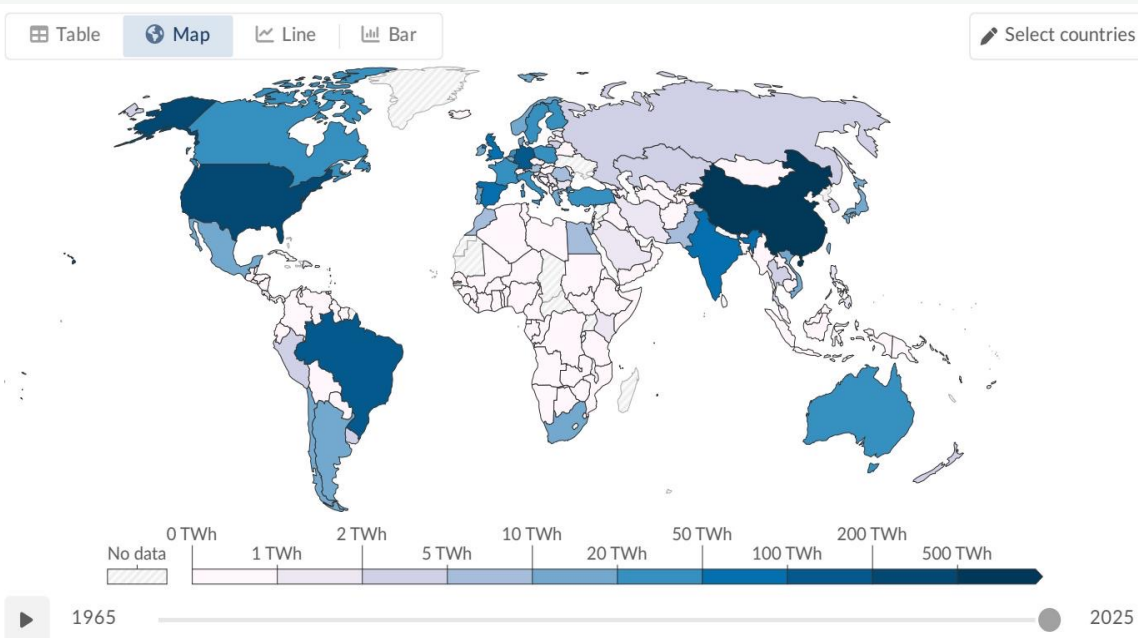


Country / Region	Renewables Share
Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%
World Average	~30%

~30% global avg · Many African/
Middle Eastern nations below 10%

Hydropower Remains the Largest Renewable Source at ~50% of Generation

Over a century of scale — China, Brazil, and Canada are top producers



Hydro growth is limited by geography and environmental constraints — wind and solar have greater untapped potential.

Source: Our World in Data – Renewable Energy (Ember 2026; EI Statistical Review 2025)

~4,500 TWh

global hydro generation (2024)

~47%

of all renewable electricity

3 leaders

China · Brazil · Canada

100+ yrs

of large-scale hydro operation

Wind and Solar Are the Fastest-Growing Energy Technologies in History

Together ~4,400 TWh — now nearly matching hydropower's 4,500 TWh output

Wind • ~2,400 TWh

~25% of renewable electricity

Onshore + offshore; strong in Europe, China, USA

Solar • ~2,000 TWh

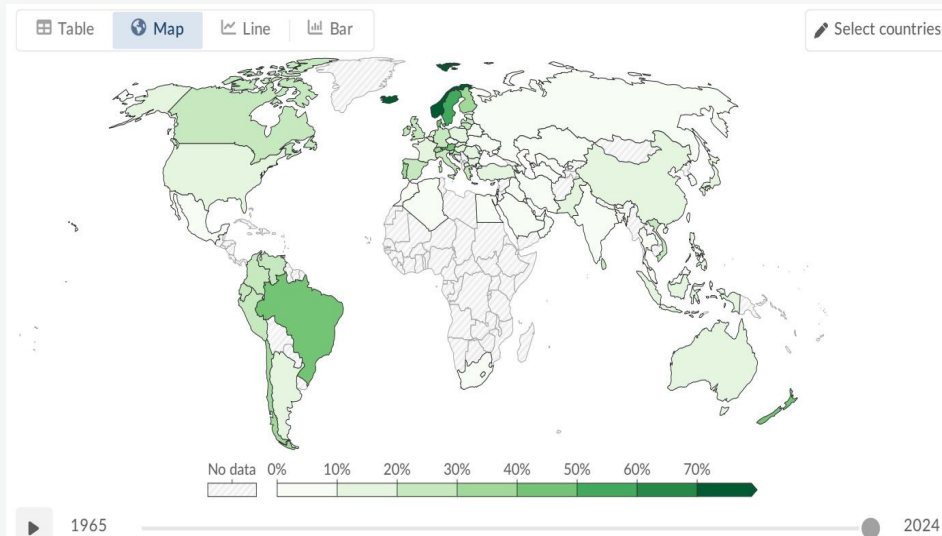
~21% of renewable electricity

Fastest-growing source; dramatic rise since 2010 — driven by cost declines & policy

Wind + Solar Combined: ~4,400 TWh ≈ Hydro (~4,500 TWh)

But growing far faster — no geographic ceiling

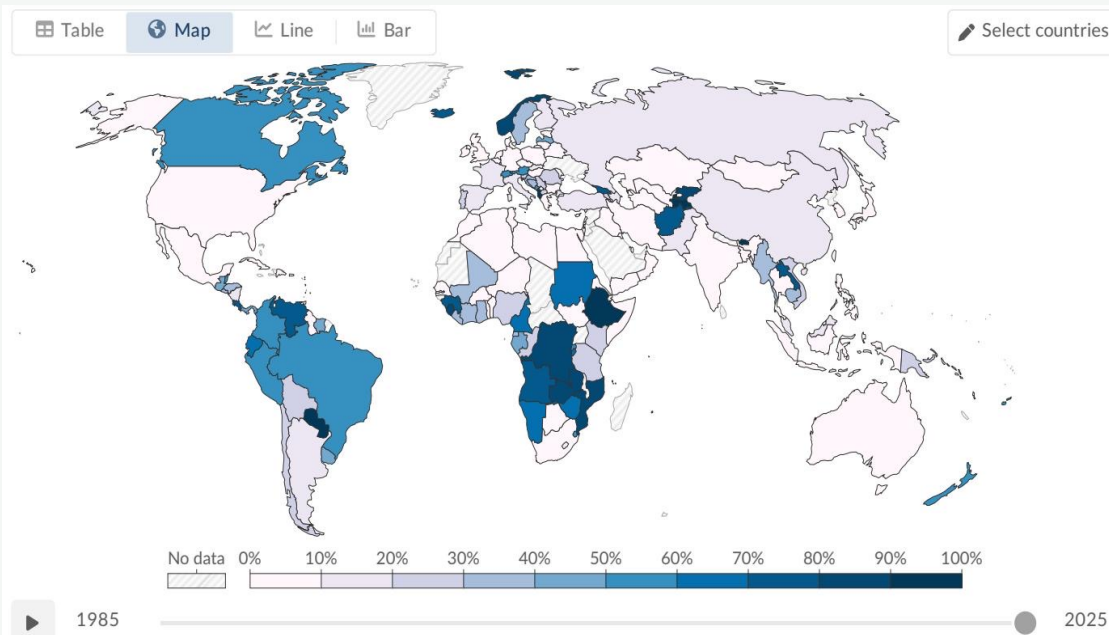
Source: Our World in Data – Renewable Energy (Ember 2026; EI Statistical Review 2025)



Wind generation by country, 2025 (Our World in Data / Ember)

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia and Latin America Lead

The energy transition is profoundly uneven — access and investment drive outcomes



Share of electricity from hydropower by country, 2025 (Our World in Data / Ember)

Source: Our World in Data – Renewable Energy (Ember 2026; EI Statistical Review 2025)

Leaders

- Norway ~98% · Brazil ~85% · Canada ~65%
- Denmark & Germany: heavy wind & solar investment
- Latin America: abundant hydropower resources

Laggards

- Saudi Arabia <2% — fossil-fuel-dependent
- Sub-Saharan Africa: minimal renewable infrastructure
- Much of Middle East below 10%

KEY TAKEAWAYS

Renewables Are Growing Fast — But the Transition Must Accelerate

- 01 ~14% of global primary energy and ~30% of global electricity now come from renewables — significant milestones, but well short of climate targets.
- 02 Hydropower dominates renewable electricity (~47%) but has limited growth potential; wind and solar are the growth engines.
- 03 Wind (~2,400 TWh) and solar (~2,000 TWh) combined now nearly match hydro output and are the fastest-growing energy technologies in history.
- 04 Stark regional gaps persist: Scandinavia and Latin America lead at 65–98%; Sub-Saharan Africa and the Middle East remain below 10%.
- 05 Transport and heating lag far behind electricity in decarbonization, keeping the overall primary energy share much lower than the electricity share.

The window to meet 1.5°C targets is narrowing — scaling wind and solar must be the top global energy priority.